Manyara Kevin Omenyi
Vladimir Borel
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Ocean Carrier

2. *Should Ms. Linn purchase the $39M capesize? Ocean Carriers uses a 9% discount rate. Make two different assumptions: OC is a US firm subject to 35% taxation. OC is located in Hong Kong, where owners of HK ships are not required to pay any tax on profits made overseas and are also exempted from paying any tax on profit made on cargo uplifted from HK.*

Input	Value
Discount rate	9.0%
Expected inflation	3.0%
US corporate tax rate (scenario a)	35.0%
Hong Kong tax rate (scenario b)	0.0%
Ship cost	$39,000,000
Payment now (t=0, Jan 2001)	10.0%
Payment in one year (t=1)	10.0%
Payment on delivery (t=2, early 2003)	80.0%
Depreciable life of ship (years)	25
Survey capex depreciation (years)	5
Scrap value at end of life	$5,000,000
Initial net working capital (2003)	$500,000
Operating cost per day (year 1)	$4,000
Opex growth above inflation	1.0%
Opex annual growth rate	4.03%
Days per year	365
Maintenance days, ages 1-5	8
Maintenance days, ages 6-10	12
Maintenance days, ages 11+	16
Charter rate year 1 (2003)	$20,000
Charter annual escalation ($day)	$200
Charter length (years)	3

Table 1: Summary of Information Provided

t (years from Jan 2001)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
Calendar year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Ship age	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Operating? (1=yes)	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Daily hire rate ($/day)	-	-	-	20	20	20	19	17	17	18	18	18	17	18	18	18	18	15
Maintenance days	0	0	0	8	8	8	8	8	12	12	12	12	12	16	16	16	16	16
Operating (hire) days	0	0	0	357	357	357	357	357	353	353	353	353	353	349	349	349	349	349
Revenue	-	-	-	**7,140**	**7,211**	**7,283**	**6,681**	**6,170**	**6,171**	**6,242**	**6,314**	**6,387**	**6,152**	**6,153**	**6,223**	**6,295**	**6,368**	**5,152**
Operating costs	-	-	-	1,460	1,519	1,580	1,644	1,710	1,779	1,851	1,925	2,003	2,083	2,167	2,255	2,346	2,440	2,538
Survey capex	-	-	-	-	-	-	300	-	-	-	-	350	-	-	-	-	-	-
Depreciation - ship	-	-	-	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560
Depreciation - survey capex	-	-	-	-	-	-	-	60	60	60	60	60	70	70	70	70	70	-
Total depreciation	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,560
EBIT	-	-	-	**4,120**	**4,133**	**4,143**	**3,477**	**2,840**	**2,773**	**2,772**	**2,769**	**2,764**	**2,439**	**2,355**	**2,339**	**2,319**	**2,297**	**1,054**
Taxes	-	-	-	1,442	1,446	1,450	1,217	994	970	970	969	967	854	824	819	812	804	369
Net operating profit after tax	-	-	-	2,678	2,686	2,693	2,260	1,846	1,802	1,801	1,800	1,797	1,585	1,531	1,520	1,508	1,493	685
+ Depreciation (non-cash)	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,560
Operating cash flow	-	-	-	**4,238**	**4,246**	**4,253**	**3,820**	**3,466**	**3,422**	**3,421**	**3,420**	**3,417**	**3,215**	**3,161**	**3,150**	**3,138**	**3,123**	**2,245**
Ship purchase payments	(3,900)	(3,900)	(31,200)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
• Survey capex	-	-	-	-	-	-	(300)	-	-	-	-	(350)	-	-	-	-	-	-
NWC balance	-	-	500	515	530	546	563	580	597	615	633	652	672	692	713	734	756	-
Change in NWC (cash)	-	-	(500)	(15)	(15)	(16)	(16)	(17)	(17)	(18)	(18)	(19)	(20)	(20)	(21)	(21)	(22)	756
After-tax salvage value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8,710
NET CASH FLOW	**(3,900)**	**(3,900)**	**(31,700)**	**4,223**	**4,231**	**4,237**	**3,504**	**3,449**	**3,405**	**3,404**	**3,401**	**3,048**	**3,196**	**3,141**	**3,129**	**3,116**	**3,101**	**11,711**

Table 2: U.S. Tax Rate (35%) (15 years)

t (years from Jan 2001)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Calendar year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Ship age	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Operating? (1=yes)	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Daily hire rate ($/day)	-	-	-	20	20	20	19	17	17	18	18	18	17	18	18	18	18	15
Maintenance days	0	0	0	8	8	8	8	8	12	12	12	12	12	16	16	16	16	16
Operating (hire) days	0	0	0	357	357	357	357	357	353	353	353	353	353	349	349	349	349	349
Revenue	-	-	-	**7,140**	**7,211**	**7,283**	**6,681**	**6,170**	**6,171**	**6,242**	**6,314**	**6,387**	**6,152**	**6,153**	**6,223**	**6,295**	**6,368**	**5,152**
Operating costs	-	-	-	1,460	1,519	1,580	1,644	1,710	1,779	1,851	1,925	2,003	2,083	2,167	2,255	2,346	2,440	2,538
Survey capex	-	-	-	-	-	-	300	-	-	-	-	350	-	-	-	-	-	-
Depreciation – ship	-	-	-	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560
Depreciation – survey capex	-	-	-	-	-	-	-	60	60	60	60	60	70	70	70	70	70	-
Total depreciation	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,560
EBIT	-	-	-	**4,120**	**4,133**	**4,143**	**3,477**	**2,840**	**2,773**	**2,772**	**2,769**	**2,764**	**2,439**	**2,355**	**2,339**	**2,319**	**2,297**	**1,054**
Taxes	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net operating profit after tax	-	-	-	4,120	4,133	4,143	3,477	2,840	2,773	2,772	2,769	2,764	2,439	2,355	2,339	2,319	2,297	1,054
+ Depreciation (non-cash)	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,560
Operating cash flow	-	-	-	**5,680**	**5,693**	**5,703**	**5,037**	**4,460**	**4,393**	**4,392**	**4,389**	**4,384**	**4,069**	**3,985**	**3,969**	**3,949**	**3,927**	**2,614**
Ship purchase payments	(3,900)	(3,900)	(31,200)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
– Survey capex	-	-	-	-	-	-	(300)	-	-	-	-	(350)	-	-	-	-	-	-
NWC balance	-	-	500	515	530	546	563	580	597	615	633	652	672	692	713	734	756	-
Change in NWC (cash)	-	-	(500)	(15)	(15)	(16)	(16)	(17)	(17)	(18)	(18)	(19)	(20)	(20)	(21)	(21)	(22)	756
After-tax salvage value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5,000
NET CASH FLOW	**(3,900)**	**(3,900)**	**(31,700)**	**5,665**	**5,677**	**5,687**	**4,721**	**4,443**	**4,375**	**4,374**	**4,371**	**4,015**	**4,049**	**3,965**	**3,948**	**3,928**	**3,905**	**8,370**

Table 3: Hong Kong Tax Rate (0%) (15 years)

Scenario	Tax rate	Life (yrs)	NPV @ 9%	IRR	Decision
Q2a - US firm, 15-yr policy	35.0%	15	–7837918	5.21%	REJECT
Q2b - Hong Kong firm, 15-yr policy	0.0%	15	–1299386	8.36%	REJECT

Table 4: Conclusion (15 years)

The NPV for both tax jurisdictions is negative, and IRR is bellow the discount rate of 9%. Therefore, the investment is not attractive in both cases on a 15 year time horizon.

3. *What do you think of the firm's policy of not operating ships over 15 years old?*

t (years from Jan 2001)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**
Calendar year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Ship age	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Operating? (1=yes)	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Daily hire rate ($/day)	-	-	-	20	20	20	19	17	17	18	18	18	17	18	18	18	18	18	15	15	15	15	15	15	15	15	15	13
Maintenance days	0	0	0	8	8	8	8	8	12	12	12	12	12	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
Operating (hire) days	0	0	0	357	357	357	357	357	353	353	353	353	353	349	349	349	349	349	349	349	349	349	349	349	349	349	349	349
Revenue	-	-	-	**7,140**	**7,211**	**7,283**	**6,681**	**6,170**	**6,171**	**6,242**	**6,314**	**6,387**	**6,152**	**6,153**	**6,223**	**6,295**	**6,368**	**5,152**	**5,212**	**5,272**	**5,332**	**5,393**	**5,115**	**5,174**	**5,233**	**5,293**	**5,354**	**4,694**
Operating costs	-	-	-	1,460	1,519	1,580	1,644	1,710	1,779	1,851	1,925	2,003	2,083	2,167	2,255	2,346	2,440	2,538	2,641	2,747	2,858	2,973	3,093	3,218	3,347	3,482	3,622	3,768
Survey capex	-	-	-	-	-	-	300	-	-	-	-	350	-	-	-	-	750	-	-	-	-	850	-	-	-	-	-	-
Depreciation – ship	-	-	-	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560
Depreciation – survey capex	-	-	-	-	-	-	-	60	60	60	60	60	70	70	70	70	70	150	150	150	150	150	170	170	170	170	170	-
Total depreciation	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,710	1,710	1,710	1,710	1,710	1,730	1,730	1,730	1,730	1,730	1,560
EBIT	-	-	-	**4,120**	**4,133**	**4,143**	**3,477**	**2,840**	**2,773**	**2,772**	**2,769**	**2,764**	**2,439**	**2,355**	**2,339**	**2,319**	**2,297**	**904**	**861**	**814**	**764**	**710**	**292**	**226**	**156**	**81**	**2**	**(635)**
Taxes	-	-	-	1,442	1,446	1,450	1,217	994	970	970	969	967	854	824	819	812	804	316	301	285	267	249	102	79	54	28	1	(222)
Net operating profit after tax	-	-	-	2,678	2,686	2,693	2,260	1,846	1,802	1,801	1,800	1,797	1,585	1,531	1,520	1,508	1,493	587	560	529	497	462	190	147	101	53	1	(413)
+ Depreciation (non-cash)	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,710	1,710	1,710	1,710	1,710	1,730	1,730	1,730	1,730	1,730	1,560
Operating cash flow	-	-	-	**4,238**	**4,246**	**4,253**	**3,820**	**3,466**	**3,422**	**3,421**	**3,420**	**3,417**	**3,215**	**3,161**	**3,150**	**3,138**	**3,123**	**2,297**	**2,270**	**2,239**	**2,207**	**2,172**	**1,920**	**1,877**	**1,831**	**1,783**	**1,731**	**1,147**
Ship purchase payments	(3,900)	(3,900)	(31,200)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
– Survey capex	-	-	-	-	-	-	(300)	-	-	-	-	(350)	-	-	-	-	(750)	-	-	-	-	(850)	-	-	-	-	-	-
NWC balance	-	-	500	515	530	546	563	580	597	615	633	652	672	692	713	734	756	779	802	826	851	877	903	930	958	987	1,016	-
Change in NWC (cash)	-	-	(500)	(15)	(15)	(16)	(16)	(17)	(17)	(18)	(18)	(19)	(20)	(20)	(21)	(21)	(22)	(23)	(23)	(24)	(25)	(26)	(26)	(27)	(28)	(29)	(30)	1,016
After-tax salvage value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3,250
NET CASH FLOW	**(3,900)**	**(3,900)**	**(31,700)**	**4,223**	**4,231**	**4,237**	**3,504**	**3,449**	**3,405**	**3,404**	**3,401**	**3,048**	**3,196**	**3,141**	**3,129**	**3,116**	**2,351**	**2,275**	**2,246**	**2,215**	**2,182**	**1,296**	**1,893**	**1,850**	**1,803**	**1,754**	**1,701**	**5,414**

Table 5: U.S. Tax Rate (35%) (25 years)

t (years from Jan 2001)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**
Calendar year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Ship age	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Operating? (1=yes)	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Daily hire rate ($/day)	-	-	-	20	20	20	19	17	17	18	18	18	17	18	18	18	18	15	15	15	15	15	15	15	15	15	15	13
Maintenance days	0	0	0	8	8	8	8	8	12	12	12	12	12	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
Operating (hire) days	0	0	0	357	357	357	357	357	353	353	353	353	353	349	349	349	349	349	349	349	349	349	349	349	349	349	349	349
Revenue	-	-	-	**7,140**	**7,211**	**7,283**	**6,681**	**6,170**	**6,171**	**6,242**	**6,314**	**6,387**	**6,152**	**6,153**	**6,223**	**6,295**	**6,368**	**5,152**	**5,212**	**5,272**	**5,332**	**5,393**	**5,115**	**5,174**	**5,233**	**5,293**	**5,354**	**4,694**
Operating costs	-	-	-	1,460	1,519	1,580	1,644	1,710	1,779	1,851	1,925	2,003	2,083	2,167	2,255	2,346	2,440	2,538	2,641	2,747	2,858	2,973	3,093	3,218	3,347	3,482	3,622	3,768
Survey capex	-	-	-	-	-	-	300	-	-	-	-	350	-	-	-	-	750	-	-	-	-	850	-	-	-	-	-	-
Depreciation – ship	-	-	-	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560	1,560
Depreciation – survey capex	-	-	-	-	-	-	-	60	60	60	60	60	70	70	70	70	70	150	150	150	150	150	170	170	170	170	170	-
Total depreciation	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,710	1,710	1,710	1,710	1,710	1,730	1,730	1,730	1,730	1,730	1,560
EBIT	-	-	-	**4,120**	**4,133**	**4,143**	**3,477**	**2,840**	**2,773**	**2,772**	**2,769**	**2,764**	**2,439**	**2,355**	**2,339**	**2,319**	**2,297**	**904**	**861**	**814**	**764**	**710**	**292**	**226**	**156**	**81**	**2**	**(635)**
Taxes	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net operating profit after tax	-	-	-	4,120	4,133	4,143	3,477	2,840	2,773	2,772	2,769	2,764	2,439	2,355	2,339	2,319	2,297	904	861	814	764	710	292	226	156	81	2	(635)
+ Depreciation (non-cash)	-	-	-	1,560	1,560	1,560	1,560	1,620	1,620	1,620	1,620	1,620	1,630	1,630	1,630	1,630	1,630	1,710	1,710	1,710	1,710	1,710	1,730	1,730	1,730	1,730	1,730	1,560
Operating cash flow	-	-	-	**5,680**	**5,693**	**5,703**	**5,037**	**4,460**	**4,393**	**4,392**	**4,389**	**4,384**	**4,069**	**3,985**	**3,969**	**3,949**	**3,927**	**2,614**	**2,571**	**2,524**	**2,474**	**2,420**	**2,022**	**1,956**	**1,886**	**1,811**	**1,732**	**925**
Ship purchase payments	(3,900)	(3,900)	(31,200)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
– Survey capex	-	-	-	-	-	-	(300)	-	-	-	-	(350)	-	-	-	-	(750)	-	-	-	-	(850)	-	-	-	-	-	-
NWC balance	-	-	500	515	530	546	563	580	597	615	633	652	672	692	713	734	756	779	802	826	851	877	903	930	958	987	1,016	-
Change in NWC (cash)	-	-	(500)	(15)	(15)	(16)	(16)	(17)	(17)	(18)	(18)	(19)	(20)	(20)	(21)	(21)	(22)	(23)	(23)	(24)	(25)	(26)	(26)	(27)	(28)	(29)	(30)	1,016
After-tax salvage value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5,000
NET CASH FLOW	**(3,900)**	**(3,900)**	**(31,700)**	**5,665**	**5,677**	**5,687**	**4,721**	**4,443**	**4,375**	**4,374**	**4,371**	**4,015**	**4,049**	**3,965**	**3,948**	**3,928**	**3,155**	**2,591**	**2,548**	**2,500**	**2,449**	**1,545**	**1,995**	**1,929**	**1,858**	**1,782**	**1,702**	**6,942**

Table 6: Hong Kong Tax Rate (0%) (25 years)

Scenario	Tax rate	Life (yrs)	NPV @ 9%	IRR	Decision
Q3 - US firm, run 25 years	35.0%	25	($7,018,743)	6.05%	REJECT
Q3 - Hong Kong firm, run 25 years	0.0%	25	$758,301	9.32%	ACCEPT

Table 7: Conclusion (25 years)

Firm	NPV, 15 years	NPV, 25 years	Value of extending
US firm (35% tax)	-$7.82M	-$6.98M	+$0.84M
Hong Kong firm (0% tax)	-$1.27M	+$0.81M	+$2.08M

Table 8: Value of running the ship for 25 instead of 15 years

Analysis for 15 years indicate that we reject investment. While the analysis on the 25 year time horizon shows that only an investment in Hong Kong with a 0% tax rate has a positive NPV and an IRR above the 9% discount rate.

We recommend changing the policy to 25 years and registering their ships in Hong Kong.